A distributed system for fining high profit areas over big taxi trip data with MognoDB and Spark

Fadhilah Kurnia Putri*, Joonho Kwon†
*Department of Big Data, †School of Computer Science and Engineering
Pusan National University, Busan, South Korea
{fadhilahkp, jhkwon}@pusan.ac.kr

***Abstract*—Due to the rapid development of internet of things (IoT) technology, traditional taxi cabs are connected through dispatchers and location systems. In major urban cities, modern taxis have equipped with GPS (Global positioning system) sensors which generate huge volumes of taxi trips. One of crucial solutions from analyzing taxi trip big data is to find high profit areas for taxi drivers. In this paper, we proposed a distributed high profit areas search system which is based on MongoDB and Spark. Our system extract information about high profit areas from raw taxi trips and store them into a MongoDB document store. If a user sends a query for requesting high profit areas, our system returns a set of high profit areas by a distributed skyline processing algorithm with Spark. Experimental results with real-work data demonstrates the feasibility and the effectiveness of our approach.**

***Keywords*-taxi trip big data; internet of things; distributed skyline processing; high profit areas;**

I. Introduction

The prevalence of internet of things (IoT) technology in transportation areas enables us to collect and analyze huge volumes of traffic big data. In many urban cities, taxi cabs have been installed with several IoT sensors such as GPS (global positioning system), DSRC (dedicated short-range communications) and cameras. Due to the IoT sensors, a taxi company could collect trip data of each vehicle and extract the patterns of taxi drivers when they provide services to passengers. A taxi driver can also obtain a lot of meaningful information which allows them to earn high profits by analyzing historical taxi trips from GPS sensors.

Figure 1 illustrates an example usage of the analyzed data to find high profit areas for taxi drivers. Assume that a taxi driver picks a passenger from area A and deliver him to area B. Then, he has four high profit areas to find a new passenger : (1) move to area H which is near subway station, (2) move to area I which is a shopping district, (3) stay in area B, or (4) move to area G which is a residential district. Each area has advantages and disadvantages due to several factors such as an average profit p, a passenger demand probability pd, average cruising time t_{cr}, an average cruising distance d_{cr}. Taxi drivers have well known high profitable areas of familiar locations by their own experiences. However, we believe that they could have advised from the analysis results of taxi trip when they happen to go unfamiliar places.

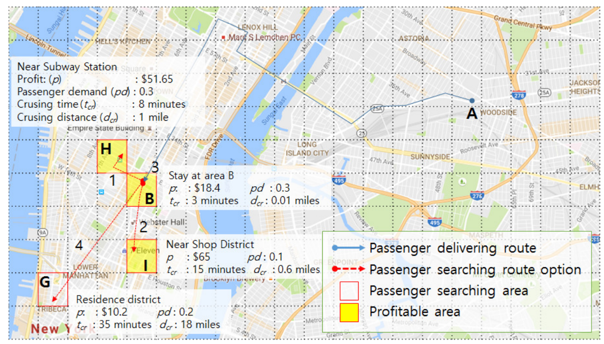

Figure 1: Where are high profit areas?

As a consequence, many researchers have been conducted research on providing the analysis results of collected GPS sensor data (= taxi trips) [1], [2], [3], [4], [5], [6], [7], [8], [9], [10]. For example, several researches have been analyzed GPS sensor data for tackling general traffic problems [11], [12], [1]. Demands and supplies of taxi cabs in Singapore are estimated to increase the efficiency of taxi companies by using real time GPS data [11]. GPS coordinates of NYC taxi trip origin and destination are analyzed to measure efficiency of taxi service using a graph-based method [12]. Zhang *et al.* [1] investigate efficient and inefficient taxi service strategies by analyzing taxi trip data. As another research direction, some researchers suggest solutions to find high profit areas based on various factors such as cruising time [2], [3], [4], maximum profit [5], cruising distance[3], or high passenger demand [6], [7], [8], [9], [10]. However, these researches considers only one or two factors although it is well known that many factors affect taxi drivers to search high profit areas.

Specialized data structures such as a spatio-temporal profitable map [2] or data mining techniques based on clustering algorithms [6], [7], [10], [9], statistical approaches (ARIMA [3], chi-square distribution [13], statistical learning [4], predictive distribution [8],probability model [14]) have been studied in the last decade. However, aforementioned approaches do not consider a distributed execution of finding high profit areas nor return a set of areas as high profit areas with multiple factors.

To address these challenge, we proposed a distributed

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system for fining high profit areas over big taxi trip data with MongoDB [15] and Spark [16]. For this purpose, first we extract information about high profit areas from raw taxi trip data by considering four factors profit, passenger demand probability, cruising time, and cruising distance in a distributed way. The extracted information are stored into a MongoDB document store. In addition, we suggest a distributed query processing algorithm for finding high profit areas based on Spark. Our system exploits an extension of skyline query processing [17] to guarantee the high profit areas as a final result are not dominated by other areas. Thus, the propose distributed query processing efficiently finds a set of high profit areas with multiple factors.

Our contributions can be summarized as follows:

- Based on MongoDB and Spark, we design and develop a distributed system for fining high profit areas over big taxi trip data.
- We suggest an extended skyline query processing algorithm for finding high profit areas with multiple factors.
- We have performed an experimental evaluation to measure the scalability and efficiency of the proposed system using real-world New York taxi trip data [18].

II. A DISTRIBUTED SYSTEM FOR FINDING HIGH PROFIT AREAS

In this section, the proposed distribution system will be discussed in detail. First, we provide the overall view of our system. Next, we explain an extraction process which converts taxi trip data to information about high profit areas. Then, we describe the detailed procedures for information extraction and query processing.

A. Overall architecture

Figure 2 illustrates an overall architecture of the proposed system. Our system mainly consists of two components: (1) a distributed information extractor and (2) a distributed query processor. The information extractor is implemented on a top of MongoDB and convert raw taxi trip data in HDFS to the information about high profit areas such as area summary and route summary explained in Section II-B. The area summary and the route summary are stored into multiple MongoDB shards. The query processor is implemented with Apache Spark core. It receives queries including a current location and a current time from users and returns a set of high profit areas to the users. Since our system consider multiple factors for high profit areas, we exploit and extend a skyline query processing which guarantee that final answers are not dominated by the other areas.

B. Distributed information extraction

Figure 3 depicts a process of the information extraction which converts raw taxi trips into specialized data structures for high profit areas.

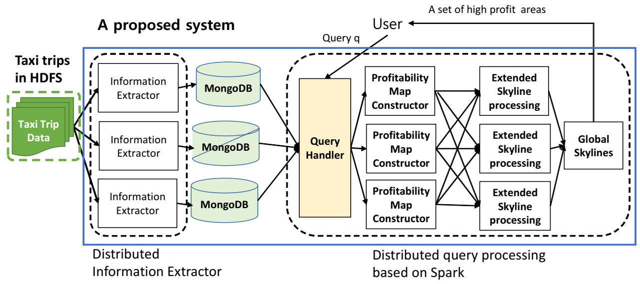

Figure 2: A distributed system architecture

Generally, taxi trip data consists of the following attributes: pickup location and time as the beginning of a trip , drop-off location and time as the end of a trip, a trip distance, fare amount, tip amount and tolls amount. An example of taxi trip data is shown at the upper part in Figure 3.

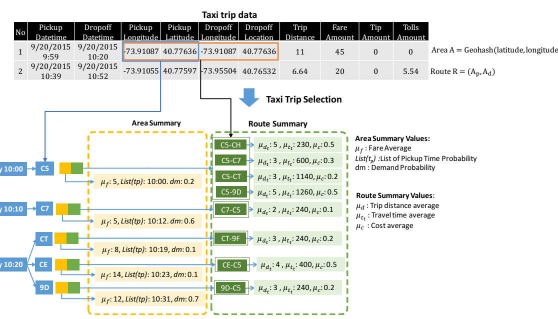

Figure 3: Information extraction

Taxi drivers plan their own routes after dropping off a passenger. Their strategies for making high profits depends on time and area. For example, it is better for a taxi driver to know the areas where he/she could pick up passengers more quickly given a certain time of a day [4]. Taxi drivers decide to move another area 'A' based on their previous experiences such as the length of typical trips which begins from the area 'A', cursing distance/time to 'A' and fuel used.

To manage information about high profit area, our system maintains a hash-based data structures which consists of thee main components: (1) a spatio-temporal key for quickly identifying information of a certain area with a given time, (2) an area summary for storing the aggregated profits an area, and (3) a route summary for providing the average expenses of the same routes which have the same pickup/drop-off locations.

During the distribute information extraction process, first we obtain a time period and an area code from a taxi trip and use them as a hash key value. Then, we identify an area summary and route summary which a taxi trip belongs to and update two summaries. After the process, the final data structures are stored into a MongoDB document store. At the bottom part of Figure 3 presents a conceptual view of data structures for high profit areas.

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C. Distributed query processing

A distributed approach for fining high profit areas consists of two main steps when a user query is given to our system as shown in the right part of Figure 2. The first step is to construct a profitability map which includes candidate high profit areas and the second step is to find the final answers of high profit areas by exploiting a skyline query processing.

The construction of a profitable map is done as follows: A user specifies his/her interest with a current location and time for requesting high profit areas. First, our system selects route summaries which has the current location as an origin area in the given time. Next, we regard all of the destination areas from these route summaries as candidates areas. Then, we access each area summary of a candidate area to obtain the expected profit values of the area. Finally, by merging route and are summaries, we could build a profitable map of candidate high profit areas. Attributes of these summaries become the factors of each candidate area. Note that the above process is executed in a distributed way with Spark.

An extended skyline processing begins by reading a partition of the profitable map which including candidate areas by. Then local skyline areas are calculated by removing all dominated areas from the partition of the profitable map at commodity servers of Spark. After that, all local profitable areas will be merged to one of Spark node to obtain global skylines of profitable areas. These global skylines are sent to the user as high profit areas.

III. Experiments

In this section, we report the performance evaluation of the proposed approach.

A. Environments

To implement our system, we used four commodity machines for a distributed processing of information extraction and queries for high profit areas. Each machine is equipped with an Intel Core 2 Duo 3.00 GHz CPU, 2GB of main memory, and runs a 64-bit Ubuntu 12.04 operating system. We implement our approach in JAVA using JDK 1.7. Spark 1.5 used as a distributed processing framework and MongoDB 3.2 is selected as a data storage for the extracted information.

We used a real taxi trip dataset collected in New York [18] city during the whole year of 2015. This dataset includes 173 million events and the could be regarded as a large scale data because a size of this dataset is about 16GB.

B. Experimental results

Information Construction

Figure 4 shows the cost breakdown of the execution time for extracting information from raw taxi trip data by varying months of the dataset. For this experiment, we fixed the number of commodity servers to 3. The first observation is that the execution time increased with the the number of months. Another observation is that the route summarization takes most of the execution time. This is because the total number of routes are much larger than total number of areas and the computation of route summaries are more complex than reading parts of raw taxi trips (taxi trip selection) and writing data structures into MongoDB (data storing).

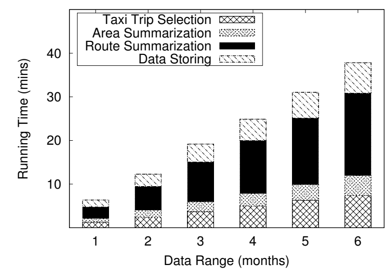

Figure 4: Performance of information extraction by varying months

Since the total size of the dataset is larger than a memory size of a typical commodity server, we devise a distributed approach for extracting information and fining high profit areas. Thus, the scalability is one of the important metrics. Figure 5 presents the results when we varied the number of commodity servers from 2 to 4. As we expected, we observe that the execution time decreases with the number of commodity severs. Another observation is that the steps for a route summary is more effectively reduced than the other process. This is because our approach treats a lot of routes effectively during the distributed process.

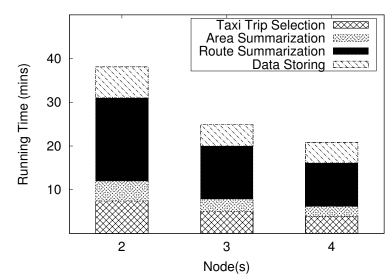

Figure 5: Distributed approach

Figure 6 shows the size comparison between raw taxi trip data and the proposed data structures which are built after the information extraction process. The size of the proposed data structures is munch lower than raw taxi trip data because we remove unused information from the raw data. The gap becomes wider with the number of months. The main reason is that duplicate areas and routes are also increased when we deal with more months of taxi trip data.

Distributed query processing

As explained in Section II-C, the distributed query processing consists of twp steps: (1) to construct a profitable map and (2) to find the final high profit area based on a skyline processing. We measured the execution time for both steps and compared our extended skyline approach with

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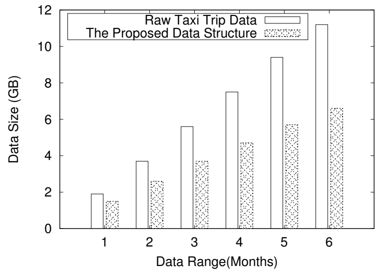

Figure 6: The size comparison

two basic skyline approaches in a distributed way. Figure 7 illustrates the execution time for a distributed search for high profit areas. Our approach performs better than previous baseline approaches which implement the distributed skyline query processing using a block-nested loop (BNL) method or a divide-and-conquer (DC) method.

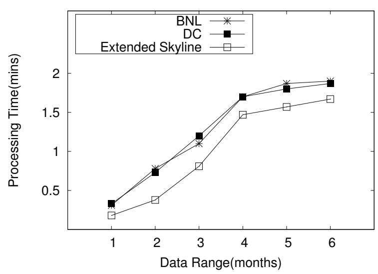

Figure 7: Profitable Areas Query Performance by Data Range

IV. Conclusion

In this paper, we investigate a problem for finding high profit areas and propose a distributed solution based on a MongoDB document store and Spark. By analyzing a huge volumes of raw taxi trips in a distributed way, we extract information about high profit areas and store them into a Mongo NoSQL database. The distributed query processing algorithm returns a skyline (subset) of high profit areas to users. We conducted experiments on real-world taxi trip big data to report effectiveness of the proposed system. For future works, we plan to study more efficient data structures for maintaining information about high profit areas and to address a fusion method to combine the historical taxi trips and real-time taxi trip data.

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